

FAJAR RAFSAN TANJUNG

Backend Developer

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PROFESSIONAL PROFILE

Backend Developer with **3+ years of hands-on experience** in backend and frontend development through an intensive scholarship program, personal projects, and teaching. Builds secure and reliable REST APIs with **Java Spring Boot, Node.js, and Golang**. Has built **six complete systems**, from hotel booking and e-commerce with online payments to flight ticketing and accounting software. Writes clean, tested, and documented code that is easy for a team to maintain. Works independently from database design and business logic to deployment with Docker. Collaborates easily with frontend teams thanks to hands-on React experience. Comfortable with Git workflows, code reviews, and clear API documentation for teammates. Taught Java and C to other students, so explains technical ideas clearly. Quick to learn new technologies. **Fresh graduate in Accounting (GPA 3.64)**, bringing extra accuracy to transactions and financial data.

TECHNICAL SKILLS

Java & Spring Boot: Java 21, Spring Boot 3/4, Spring Security, Spring Data JPA, Hibernate, Maven — used in **5 end-to-end systems**

Node.js & Express: JavaScript (ES Modules), Express 5, Prisma ORM, JWT auth, Joi, Winston, Jest, Supertest — **AniStream REST API with 49 endpoints**

Go (Golang): Go 1.22, net/http routing, database/sql with MySQL, golang-migrate (embedded SQL migrations), layered handler–service–repository design, standard project layout (cmd/, internal/), Docker multi-stage builds — **REST API service with CRUD, pagination and health checks**

Microservices & Messaging: Spring Cloud (Eureka Service Discovery, API Gateway, OpenFeign), database-per-service, transactional domain events, RabbitMQ (topic exchange, dead-letter queues), Redis caching, WebSocket/STOMP, scheduled jobs

API & Security: RESTful API design, JWT in httpOnly cookies, refresh-token rotation, role-based access control, Google OAuth2, email OTP, Argon2/BCrypt, rate limiting (Bucket4j, Redis), webhook signature verification, centralised exception handling, OpenAPI/Swagger, Postman

Databases: PostgreSQL, MySQL, SQL, Redis, Flyway & Prisma migrations, relational schema design, normalisation, indexing, query optimisation, transactions, pessimistic locking, data integrity

DevOps & Tools: Docker, Docker Compose (multi-service stacks with PostgreSQL, Redis, RabbitMQ), Git, GitHub, Maven, npm/pnpm, Spring Actuator

Testing & Quality: JUnit 5, Mockito, Spring Boot Test, MockMvc, Jest, Supertest — **240+ automated tests across 5 Spring Boot projects**

Payments & Integrations: Midtrans (Snap & Virtual Account), Xendit, RajaOngkir, Clouinary, SMTP mail, Apache POI, iText / OpenPDF

Frontend (supporting): React 19, TypeScript, Tailwind CSS 4, Vite, Vitest, Thymeleaf — used to build and test the clients of the APIs developed

Fundamentals: OOP, SOLID, MVC, dependency injection, design patterns, data structures & algorithms, clean code, C, Python (basic)

Languages: Indonesian (native) · English (technical reading & writing)

FEATURED PROJECTS

Roomly — Hotel Booking Platform (Event-Driven Microservices)

Jul 2026

Java 21 · Spring Boot 3.5 · Spring Cloud · PostgreSQL 16 · Redis 7 · RabbitMQ · Docker Compose · Midtrans · React 19 · TypeScript | **Backend:** github.com/fajarrafsan/REST-API-Hotel-Booking · **Frontend:** github.com/fajarrafsan/RoomlyHotel

- Built an **11-service Spring Cloud system** — 9 domain services plus API gateway and Eureka discovery in a 12-module Maven build — exposing **160 endpoints across 49 controllers** over **32 PostgreSQL tables**, all from one docker-compose.
- Wired **26 domain event types** through 2 RabbitMQ exchanges, 4 queues and 10 bindings with dead-letter handling, publishing only after database commit so invoice and notification work never blocks a booking response.
- Verified Midtrans webhooks by **HMAC SHA-512**, revalidating amount, currency and ownership server-side; refunds stay idempotent through row-level locks.
- Stopped double-booking with pessimistic write locks across the **5 booking tables**, and froze each booking's refund policy at creation so later edits cannot rewrite history.

GlowMarket — Gold Jewellery E-Commerce with a Built-In Accounting Engine

Sep 2025

Java 21 · Spring Boot 4 · PostgreSQL · Flyway · Xendit · RajaOngkir · WebSocket/STOMP · Clouinary · React 19 · Vite 7 · Tailwind 4 | **Backend:** github.com/fajarrafsan/REST-API-GLOWMARKET · **Frontend:** github.com/fajarrafsan/GLOWMARKET · **Demo:** projek-react.vercel.app

- Delivered **142 REST endpoints across 31 controllers and 24 business domains** — products, orders, payments, chat, accounting, loyalty, returns, shipping, notifications, analytics.
- Built a **double-entry accounting engine** over 31 tables: sales, refunds and stock adjustments post balanced journal entries through one service that rejects any unbalanced entry, surfaced as **4 reports** — general journal, ledger, P&L and balance sheet — each with Excel export.
- Integrated Xendit with a **10-minute reconciliation job** that recovers dropped webhooks; a confirmed payment posts journals and awards loyalty points atomically.
- Added real-time customer–admin chat over STOMP/WebSocket across **8 chat endpoints** with presence and offline bot replies, plus RajaOngkir rates with stored-tariff fallback.

TiketKilat — Flight Booking & E-Ticketing Platform

Jul 2025

Java 21 · Spring Boot 3.4 · PostgreSQL · Spring Security · Midtrans Snap · iText + ZXing · Apache POI · React 19 · TypeScript 5.9 · Tailwind 4 | **Backend:** github.com/fajarrafsan/TiketKilatBE · **Frontend:** github.com/fajarrafsan/TiketKilatFE

- Built a **5-stage booking flow** across **37 endpoints** and 9 PostgreSQL tables: flight search → passenger data with ID upload → Midtrans Snap payment → seat selection → QR-coded PDF e-ticket by email.
- Took payment status only from signature-verified webhooks, cross-checked server-side on **4 fields** — ownership, nominal, currency and fraud status — never from the browser redirect.

- Automated the lifecycle with **6 scheduled jobs**: a 15-minute payment deadline, OTP and token expiry, flight status and seat-availability sync.

AniStream — Streaming Platform with Dual-Source Catalogue Aggregation

Mar 2026

Node.js (ESM) · Express 5 · Prisma 6 · PostgreSQL 15 · Redis 7 · Google OAuth2 · Argon2 · Cloudinary · React 19 · Vite 8 | **Demo:** [anistream-fe-nine.vercel.app](#) · **Backend:** [github.com/fajarrafsan/REST-API-ANISTREAM-BE](#) · **Frontend:** [github.com/fajarrafsan/ANISTREAM-FE](#)

- Structured **50 endpoints across 9 route modules** through a route → controller → service → repository → mapper pipeline, with Joi validation and Winston logging.
- Wrote a **6-tier title-matching algorithm** (exact through Dice coefficient) to reconcile two upstream catalogues that name the same records differently.
- Implemented single-flight Redis caching with per-key TTLs that falls back to the source API when Redis is unavailable, over a **9-model Prisma schema** with 11 migrations and 15 indexes.

AURUMKALA — Gold Price Monitoring API — 2026 · Java 21 · Spring Boot 3.5 · PostgreSQL/H2 · Scheduled Jobs · OpenAPI | Tracks official Antam and UBS gold and buyback prices per brand and denomination, keeping a historical snapshot of every change. A scheduled job refreshes prices **three times daily, Monday–Saturday**, behind a cache guard so source sites are never hammered, with **8 endpoints** across admin re-sync, CSV export and a public OpenAPI-documented /gold-price endpoint. · [github.com/fajarrafsan/AURUMKALA](#)

SIA — Accounting Information System — Jan 2025 · Java · Spring Boot · Spring Security · MySQL · Thymeleaf | Digitised financial record-keeping across **10 entities and 45 endpoints in 9 controllers**, with a chart of accounts, journals, general ledger and financial statements on correct double-entry logic drawn from my Accounting coursework; secured with Spring Security and served through **26 Spring MVC + Thymeleaf views**. · [sia-sistem-informasi-akuntansi.onrender.com](#) · [github.com/fajarrafsan/SIA-SISTEM_INFORMASI_AKUNTANSI](#)

SHOPIFY — Console Store System in C — Dec 2023 · C · No framework | A retail store system written entirely in C with no framework — **4,771 lines across 127 functions, 26 menu options over 9 arrow-key screens**, with every record persisted to **11 delimited text files** through raw file I/O. The fundamentals layer beneath everything above it: data handling, program flow and persistence done by hand rather than by an abstraction. · [github.com/fajarrafsan/Shopify-Console-C](#)

FRONTEND CAPABILITY — SUPPORTING STRENGTH

- Builds React 19 and TypeScript frontends to test and consume the APIs developed, including a **27-route, 350-component** commerce and admin console and a **17-route** hotel booking dashboard.
- Supports API contract design, defect triage between client and server, and collaboration with frontend teams.

PROFESSIONAL EXPERIENCE

Java Fundamentals Instructor

Sep 2025 – Dec 2025

Program Beasiswa PUB · Universitas Nasional Pasim · Bandung

- Officially appointed to lead a Java fundamentals class of **5 students over 4 months**, delivering the full path from basic logic and control structures through object-oriented programming to Web MVC architecture.
- Took all 5 from lectures and quizzes to a working Spring Boot and Thymeleaf web application, assessed across **3 components** — final project, presentation and code review.

C Programming Mentor

Sep 2024 – Mar 2025

Program Beasiswa PUB · Universitas Nasional Pasim · Bandung

- Mentored PUB scholarship participants in C fundamentals over **7 months**, covering **5 core areas**: pointers, manual memory management, arrays and structs, control flow, and file I/O.

Accounting Teaching Assistant

Sep 2024 – Jun 2026

Universitas Nasional Pasim · Bandung

- Trusted by the lecturer to run tutorial and problem-session classes for **3 courses** — Introductory Accounting I & II and Public Sector Accounting — across **30 students**, preparing supplementary material, working through case studies, and setting, distributing and marking weekly quizzes.
- Marked **weekly quizzes**, assignments and examinations for all 30 students under strict accuracy requirements, and held out-of-hours consultation sessions for students needing additional support.

EDUCATION

Bachelor of Accounting (S1 Akuntansi)

2023 – 2026

Universitas Nasional Pasim · Bandung

GPA 3.64 / 4.00 · Full PUB Scholarship recipient (competitive national screening), 2023–2026

Double-entry logic, ledger structure and audit trails are not abstractions I had to look up — they show up directly in GlowMarket's accounting engine and in SIA.

CERTIFICATIONS & TRAINING

Program Beasiswa PUB (Certificate of Appreciation): Advanced Front-End (React), Jul 2026 · Fundamental Front-End (React), Jan 2026 · Advanced Back-End (Java), May 2025 · Fundamental Back-End (Java), May 2025 · GitHub, May 2025 · HTML, May 2025 · Data Structure, Jun 2024 · Database, Jun 2024 · Logic and Algorithm, Sep 2023

ACHIEVEMENTS & RECOGNITION

- Full PUB Scholarship** — selected through competitive national screening; funded 3+ years of programming bootcamp training (9 stages, 16 sessions each) alongside the Accounting degree, assessed by quizzes, handwritten coding midterms with no computer, and a built project as the final exam.
- Trusted to teach two languages by the university** — Java Fundamentals Instructor (Sep–Dec 2025) and C Programming Mentor (Sep 2024–Mar 2025) at Program Beasiswa PUB, alongside an Accounting Teaching Assistant appointment covering 30 students across three courses.
- Portfolio:** <https://fajarrafsan.my.id/> · **GitHub:** <https://github.com/fajarrafsan> — **9 end-to-end systems**, all public, documented, and runnable.